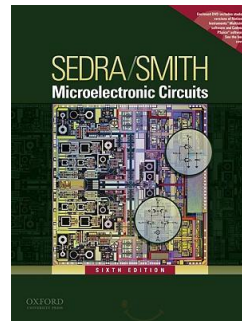


Partie 1 : Dispositifs et Circuits de base

• 4 : BJT

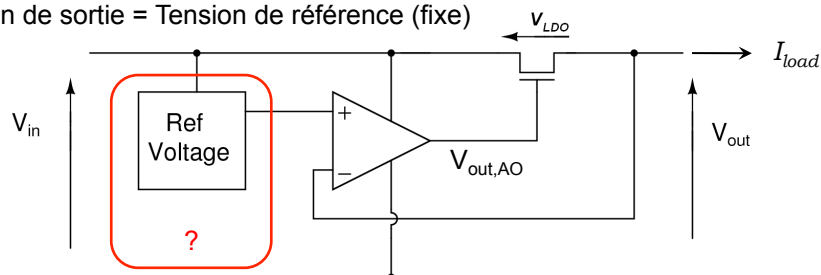
Transistor bipolaire



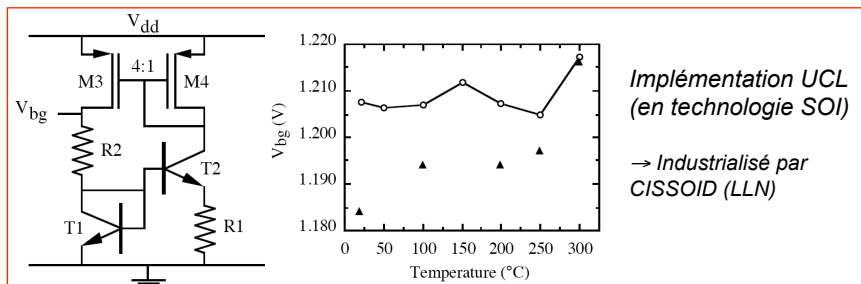
Chapter 4

Introduction : Régulateur LDO (Linear Drop Out)

Tension de sortie = Tension de référence (fixe)



Ex.:

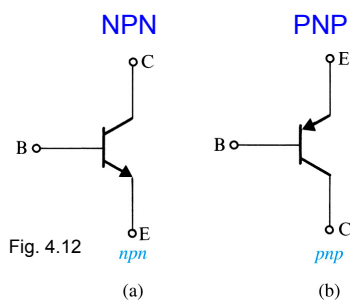


Plan du cours

- 4.1 Structure du transistor BJT
- 4.2 Caractéristiques courant – tension
- 4.3 Circuits BJT DC
- 4.4 Amplificateurs BJT en 'grand-signal'
- 4.5 Fonctionnement et modèles 'petit-signal'
- 4.6 Configurations d'amplificateurs BJT
- 4.7 Polarisation
- 4.8 Amplificateurs BJT discrets
- 4.9 Effets supplémentaires

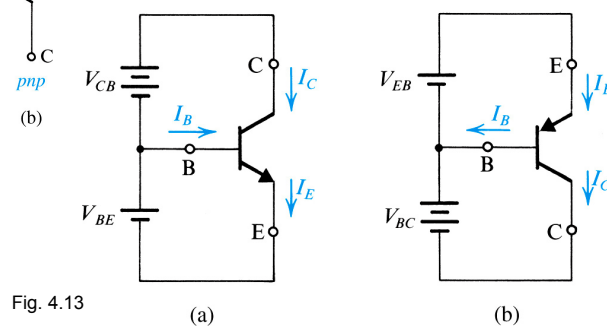
4.2 Caractéristiques I - V du BJT

Symboles :



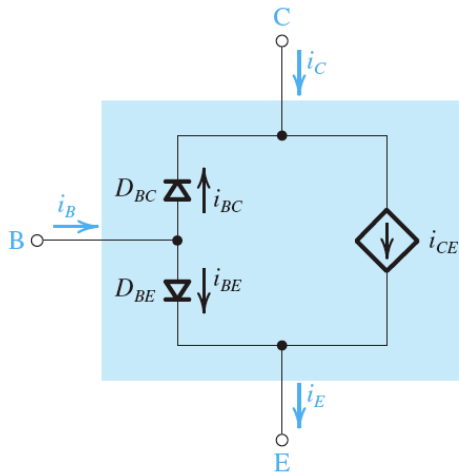
Conventions

C = Collecteur
B = Base
E = Emetteur



Modèle Ebers-Moll

Grand-signal, DC : NPN



Equations SPICE

$$i_{BE} = \frac{I_S}{\beta_F} (e^{v_{BE}/n_F V_T} - 1)$$

$$i_{BC} = \frac{I_S}{\beta_R} (e^{v_{BC}/n_R V_T} - 1)$$

$$i_{CE} = I_S (e^{v_{BE}/n_F V_T} - e^{v_{BC}/n_R V_T}) \cdot \left(1 - \frac{v_{BC}}{V_A}\right)$$

(n_F , n_R et V_A : facteurs de non-idéalités)

Courbes I-V

• $i_C - v_{BE}$

En régime actif :

$$v_{BE} > 0, \quad v_{BC} < 0$$

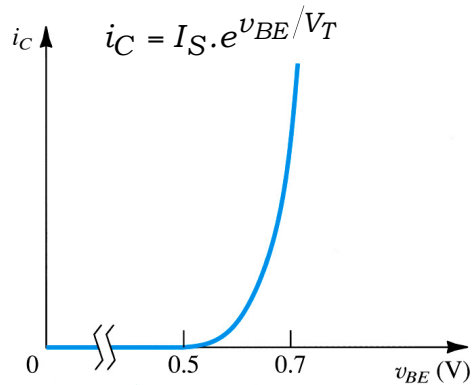


Fig. 4.15

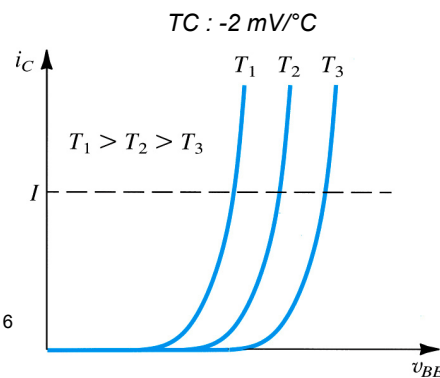
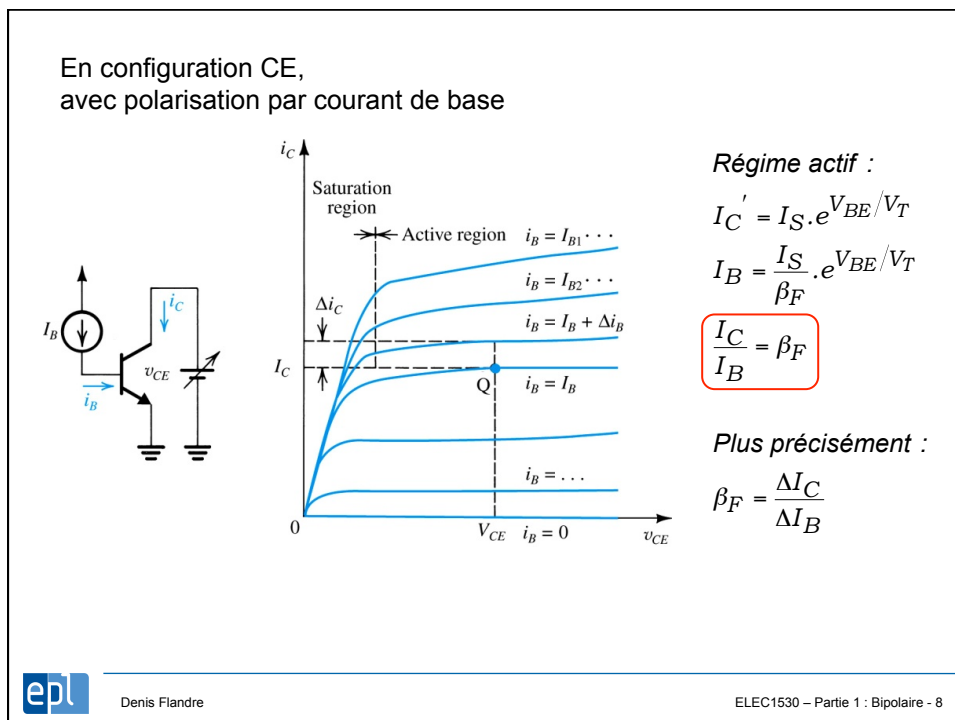
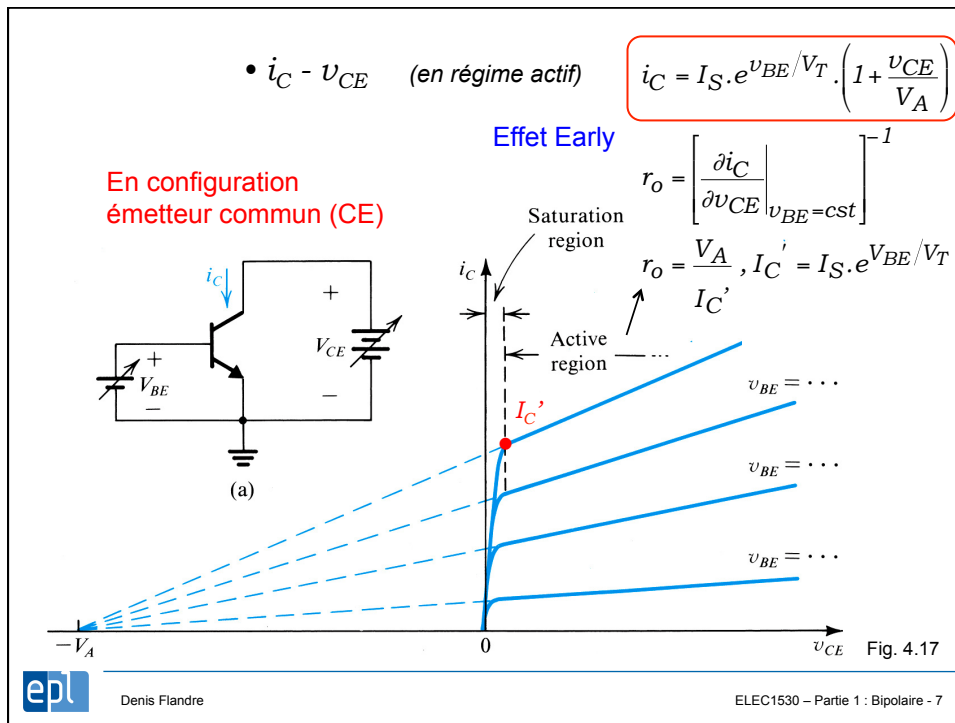


Fig. 4.16



Circuits équivalents en régime actif ($v_{ce} \geq 0.3 \text{ V}$)

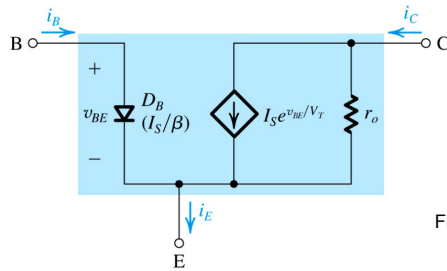
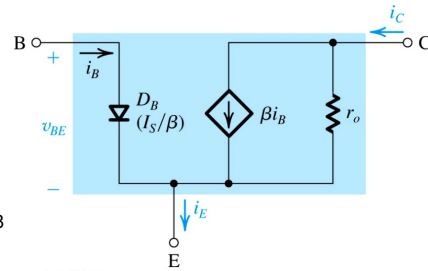


Fig. 4.18



(a)

si r_o négligée :

$$\beta = \beta_F$$

$$i_C = I_S \cdot e^{v_{BE}/V_T}$$

$$i_B = \frac{i_C}{\beta}$$

$$i_E = i_C + i_B = \frac{1}{\alpha} i_C$$

$$\alpha = \frac{\beta}{\beta + 1}$$

$$\beta = \frac{\alpha}{1 - \alpha}$$

$$i_C = \alpha \cdot i_E$$

$$i_C = \beta \cdot i_B$$

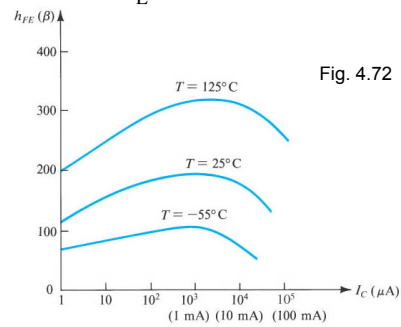
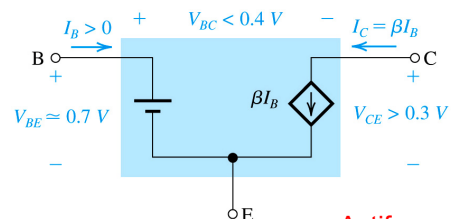
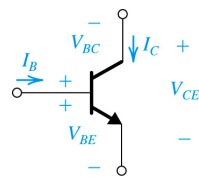


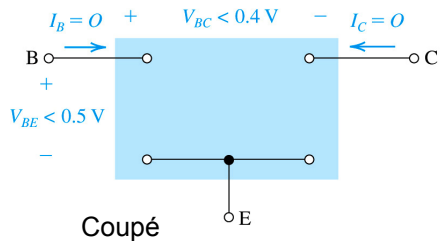
Fig. 4.72

Autres régions de fonctionnement du BJT NPN

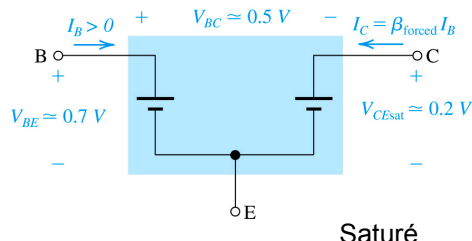
Tab. 4.3



**Actif
(Inverse)**



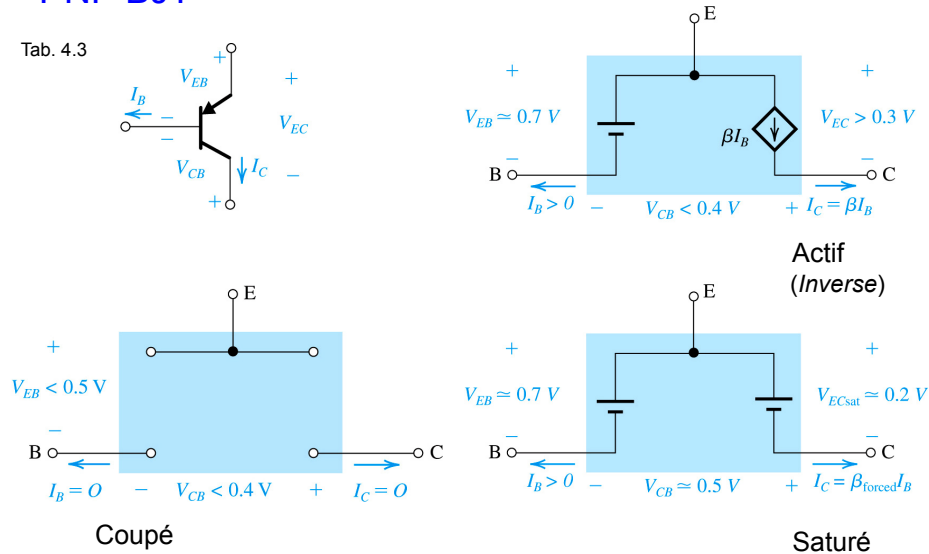
Coupé



Saturé

PNP BJT

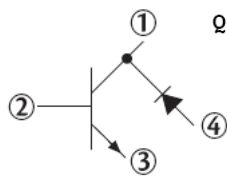
Tab. 4.3



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ELEC1530 – Partie 1 : Bipolaire - 11

SPICE avec BJT de la Sedra_Lib



`QXXXXXXX 1 2 3 <4> mname <area> <OFF> <IC=vbe, vce>
<TEMP=t> <M=# of parallel devices>`

- | | |
|---|---|
| 1 | Collector Node |
| 2 | Base Node |
| 3 | Emitter Node |
| 4 | Substrate Node. Optional. If unspecified, ground is used. |

•Model for 2N3904 NPN BJT (from Eval library in Pspice)

```
.model Q2N3904 NPN(Is=6.734f Xti=3 Eg=1.11 Vaf=74.03 Bf=416.4 Ne=1.259
+ Ise=6.734f Ikf=66.78m Xtb=1.5 Br=.7371 Nc=2 Isc=0 Ikr=0 Rc=1
+ Cjc=3.638p Mjc=.3085 Vjc=.75 Fc=.5 Cje=4.493p Mje=.2593 Vje=.75
+ Tr=239.5n Tf=301.2p Itf=.4 Vtf=4 Xtf=2 Rb=10)
```

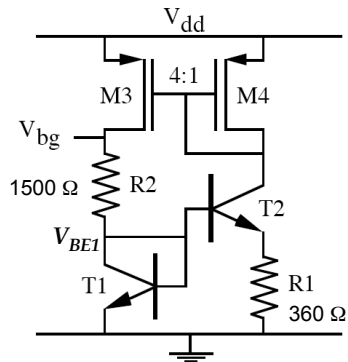


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4.3 Circuits BJT en DC

Exemple 1 :
Référence 'Bandgap'



$$I_{C1} = I_S \cdot e^{V_{BE1}/V_T} \quad I_{C2} = I_S \cdot e^{V_{BE2}/V_T}$$

$$V_{BG} = V_{BE1} + R_2 \cdot I_{C1}$$

$$V_{BE1} = V_{BE2} + R_1 \cdot I_{C2} \quad I_{C1} = 4 \cdot I_{C2}$$

$$\Rightarrow e^{(V_{BE1} - V_{BE2})/V_T} = 4$$

$$\Rightarrow V_{BE1} - V_{BE2} = 1,38 V_T = R_1 \cdot I_{C2}$$

A 25°C, $\nearrow$

$V_T \approx 26mV$ $\nearrow$

$36mV = R_1 \cdot I_{C2} \Rightarrow I_{C2} = 100\mu A$ $\nearrow$

$V_{BE1} \approx 0,7V$ $\searrow$

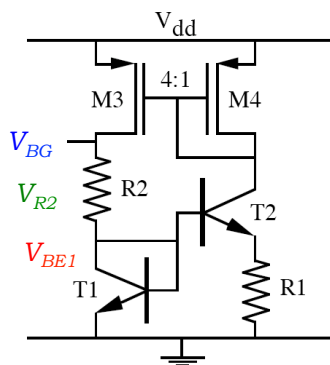
$I_{C1} = 400\mu A$ $\searrow$

$V_{BG} = 1,3V$ $=$



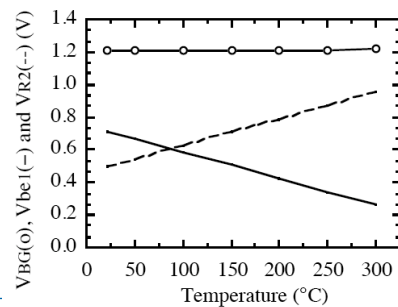
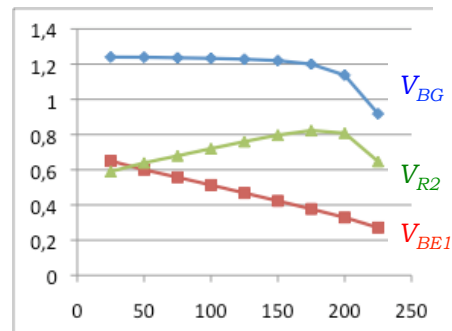
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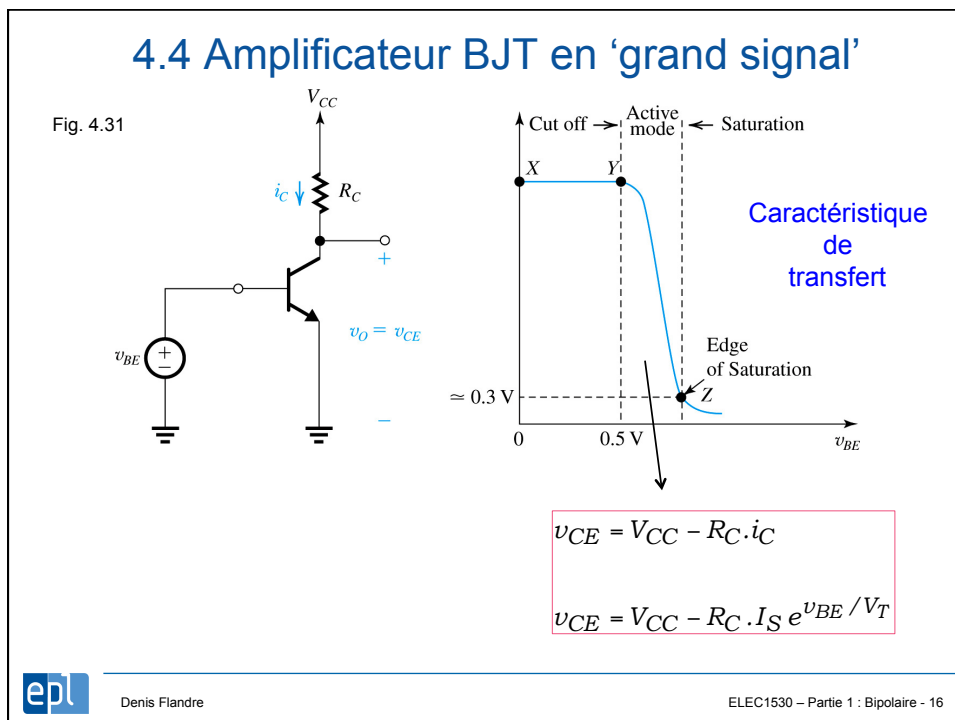
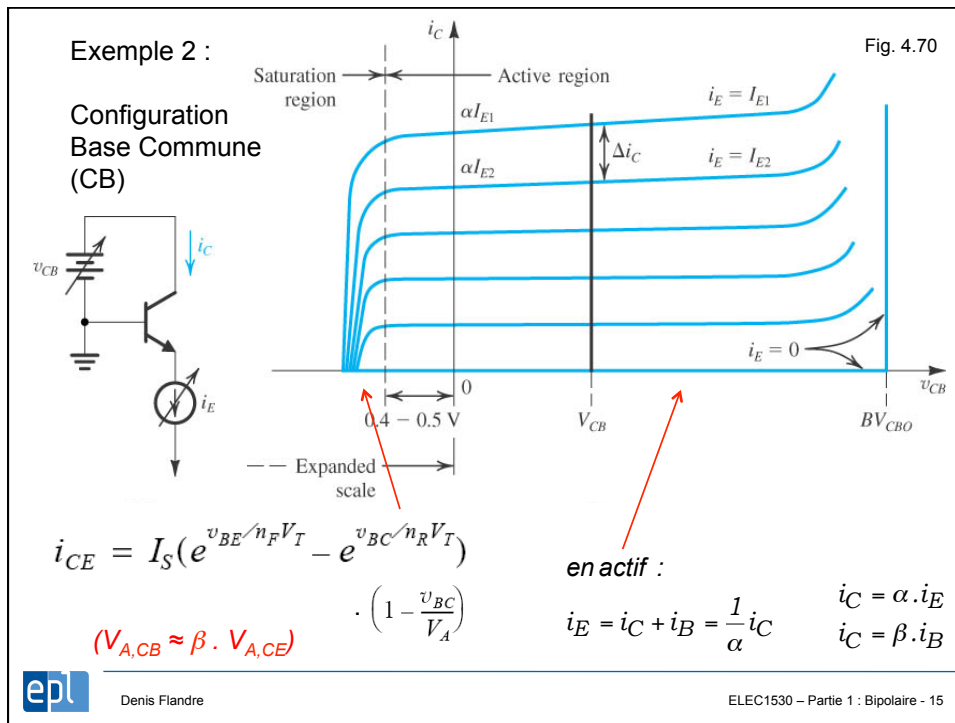
```
VDD 9 0 3
M3 4 3 9 9 PMOS0P5 L=5U W=80U
M4 3 3 9 9 PMOS0P5 L=5U W=20U
R2 4 5 1.5k
Q1 5 5 0 Q2N3904
R1 6 0 360
Q2 3 5 6 Q2N3904

.temp 25 50 75 100 125 150 175 200 225
.OP
```



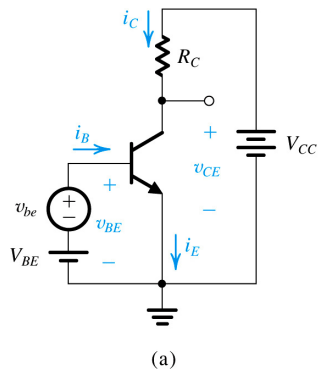
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4.5 Fonctionnement et modèles 'petit-signal'

Fig. 4.36



- Point de polarisation DC

$$I_C = I_S \cdot e^{V_{BE}/V_T} \quad I_B = I_C / \beta$$

$$V_{CE} = V_{CC} - R_C \cdot I_C > 0,7 \text{ V} \dots (\text{actif})$$

- Signal variationnel

$$v_{BE} = V_{BE} + v_{be} \Rightarrow$$

$$i_C = I_S \cdot e^{(V_{BE} + v_{be})/V_T}$$

$$= I_S \cdot e^{V_{BE}/V_T} \cdot e^{v_{be}/V_T} = I_C \cdot e^{v_{be}/V_T}$$

$$\approx I_C \cdot \left(1 + \frac{v_{be}}{V_T}\right) \quad \text{si } v_{be} \ll V_T$$

$$= I_C + \frac{I_C}{V_T} \cdot v_{be}$$

DC

$\frac{I_C}{V_T} \cdot v_{be}$
 i_c
ac

Approximation petit-signal

$$i_c = \frac{I_C}{V_T} v_{be} = g_m \cdot v_{be}$$

$$g_m = \frac{I_C}{V_T}$$

$$g_m = \left. \frac{\partial i_C}{\partial v_{BE}} \right|_{i_C = I_C}$$

Transconductance petit-signal, g_m

Unités : Mhos (Ω^{-1}) ou Siemens (S)

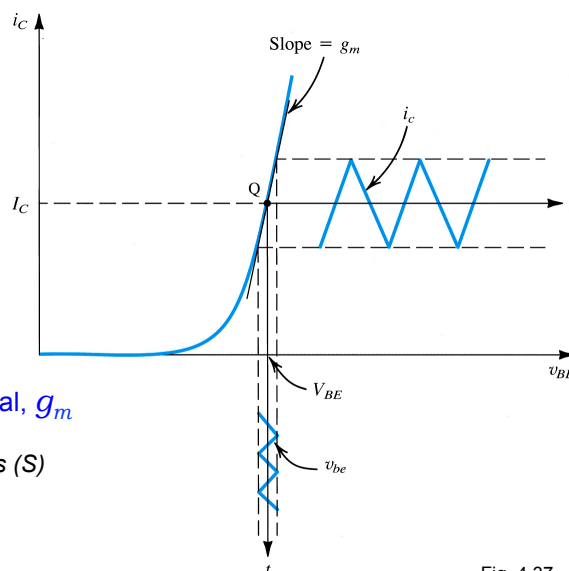


Fig. 4.37

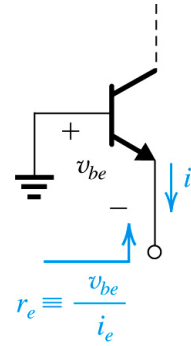
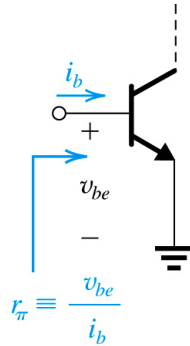
Résistances (d'entrée) de base et d'émetteur

$$i_C = I_C + \frac{I_C}{V_T} \cdot v_{be}$$

$$i_B = \frac{i_C}{\beta} = \frac{I_C}{\beta} + \frac{1}{\beta} \frac{I_C}{V_T} \cdot v_{be}$$

$$i_b = \frac{1}{\beta} \frac{I_C}{V_T} \cdot v_{be} = \frac{g_m}{\beta} \cdot v_{be}$$

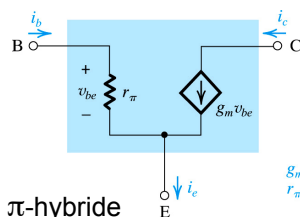
$$r_\pi \equiv \frac{v_{be}}{i_b} =$$



$$i_E = \frac{i_C}{\alpha} = \frac{I_C}{\alpha} + \frac{i_c}{\alpha} \Rightarrow i_e = \frac{i_c}{\alpha} = \frac{1}{\alpha} \frac{I_C}{V_T} \cdot v_{be} = \frac{I_E}{V_T} \cdot v_{be} \Rightarrow r_e \equiv \frac{v_{be}}{i_e} =$$

$$v_{be} = i_b \cdot r_\pi = i_e \cdot r_e \Rightarrow r_\pi = (i_e / i_b) \cdot r_e = (\beta + 1) \cdot r_e$$

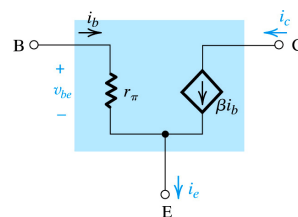
Circuits équivalents 'petit-signal'



Modèle π -hybride

$$g_m = I_C / V_T$$

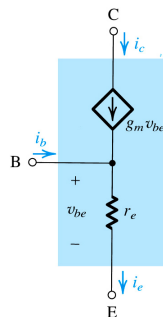
$$r_\pi = \beta / g_m$$



Modèle en T

$$i_b = \frac{i_c}{\beta}$$

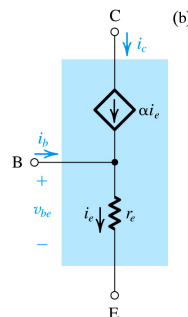
$$i_e = i_b + i_c$$



(a)

$$g_m = I_C / V_T$$

$$r_e = \frac{V_T}{I_E} = \frac{\alpha}{g_m}$$

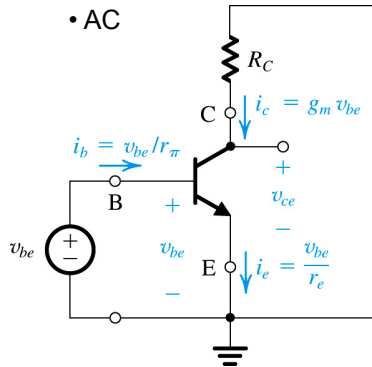


(b)

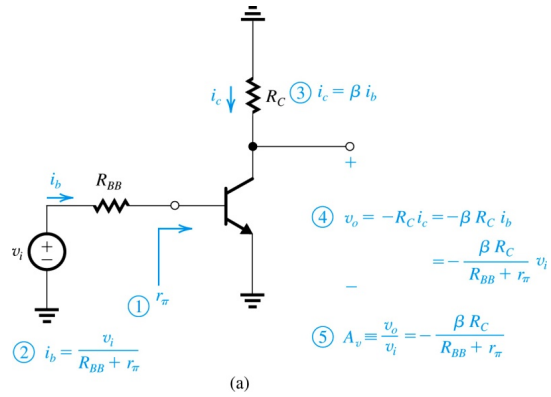
$$r_e \approx r_\pi / \beta$$

Gain ac

- DC : I_C
- AC

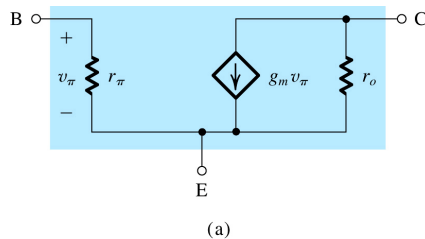


$$A_v \equiv \frac{v_{ce}}{v_{be}} =$$



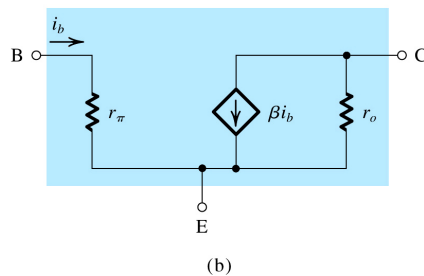
$$A_o \equiv \frac{v_o}{v_i} =$$

...avec résistance de sortie finie



$$i_C = I_S \cdot e^{v_{BE}/V_T} \left(1 + \frac{v_{CE}}{V_A} \right)$$

$$r_o = \left[\frac{\partial i_C}{\partial v_{CE}} \bigg|_{v_{BE}=cst} \right]^{-1}$$



$$r_o = \frac{V_A}{I_C'} \approx \frac{V_A}{I_C}$$

$$v_o = -g_m \cdot v_{be} \cdot (R_C \parallel r_o)$$

4.6 Configurations d'amplificateurs BJT

3 configurations de base

Représentations
'petit-signal'

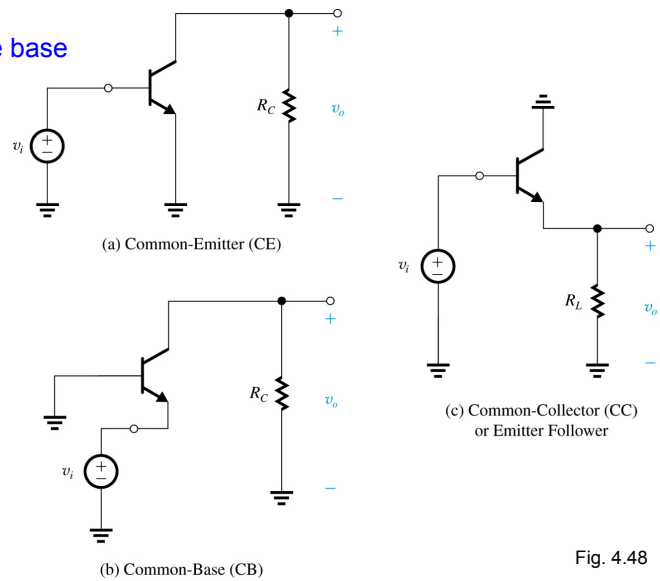


Fig. 4.48

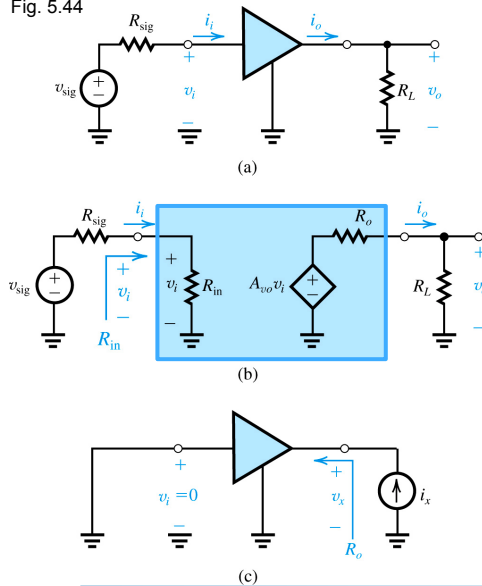


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ELEC1530 – Partie 1 : Bipolaire - 25

Méthodologie : caractérisation des amplificateurs

Fig. 5.44



$$R_{in} =$$

$$v_i =$$

$$A_{vo} =$$

$$R_o =$$

$$v_o =$$

$$A_v =$$

$$G_v =$$

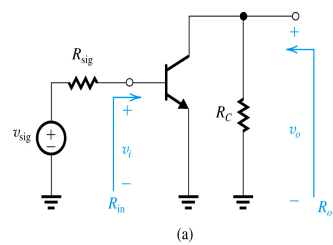
$$G_v =$$



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ELEC1530 – Partie 1 : Bipolaire - 26

Amplificateur en émetteur commun (ac)



$$R_{in} =$$

$$A_{\nu 0} =$$

$$R_o =$$

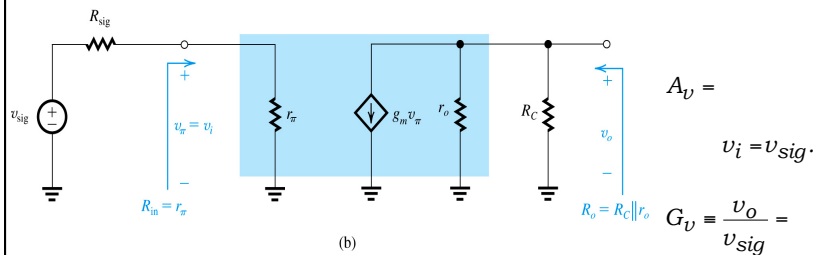
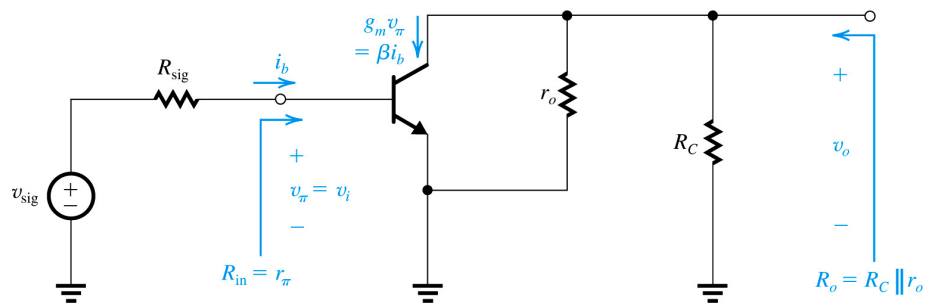


Fig. 4.50

... analyse directe

Fig. 4.51



$$r_e = \frac{\alpha}{g_m}$$

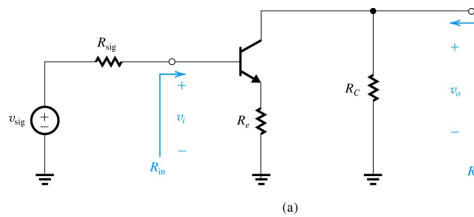
$$r_{\pi} = \beta / g_m$$

$$v_o = -g_m v_\pi (R_C \parallel r_o)$$

$$A_v = -g_m \cdot (R_C \parallel r_o) =$$

$$G_v = A_v \frac{r_\pi}{R_{sig} + r_\pi} =$$

... avec résistance d'émetteur ('degeneration')



$$\frac{v_{be}}{v_i} =$$

$$R_{in} =$$

=

≈

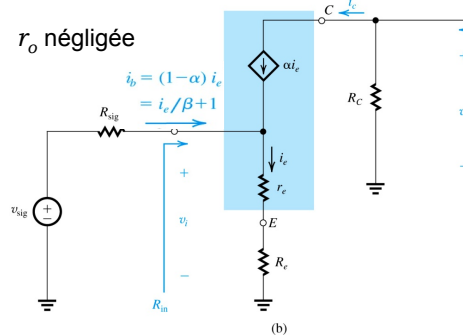
$$A_{vo} = \frac{v_o}{v_i} =$$

...

= "negative feedback" (Chap. 9)
⇒ Amélioration de performances
au prix d'un gain plus faible !

Fig. 4.52

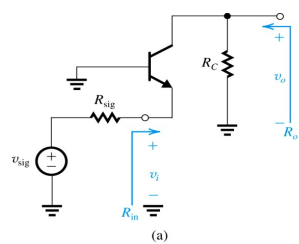
r_o négligée



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ELEC1530 – Partie 1 : Bipolaire - 29

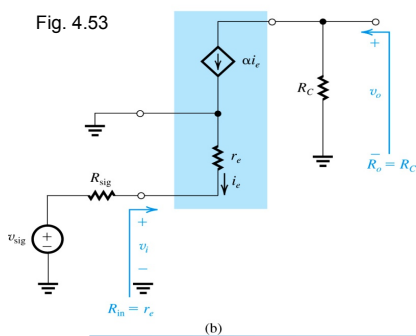
Amplificateur en base commune (ac)



$$R_{in} =$$

$$A_{vo} = \frac{v_o}{v_i} =$$

Fig. 4.53



$$R_o =$$

$$\frac{v_i}{v_{sig}} =$$

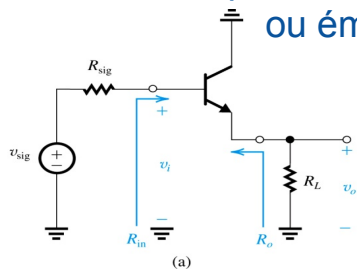
$$G_v = \frac{v_o}{v_i} \frac{v_i}{v_{sig}} =$$



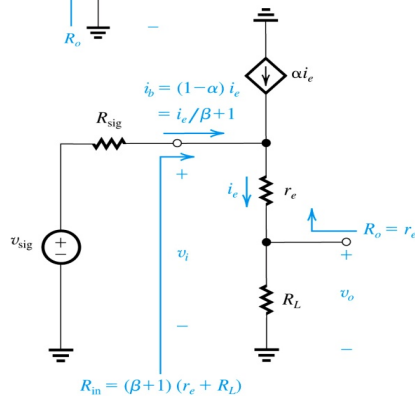
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ELEC1530 – Partie 1 : Bipolaire - 30

Amplificateur en collecteur commun ou émetteur suiveur (ac)



(a)



$$R_{in} =$$

$$R_o =$$

$$A_{vo} = \frac{v_o}{v_i} =$$

$$G_v = \frac{v_i}{v_{sig}} \times A_v \approx$$



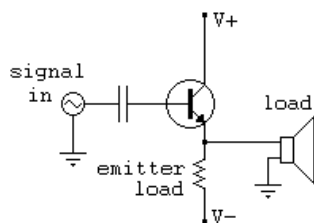
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ELEC1530 – Partie 1 : Bipolaire - 31

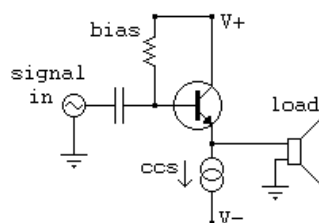
Exemple : amplificateur pour écouteur

Spécifications : 32 - 600 Ω , 80 - 450 mW (sortie)

http://gilmore2.chem.northwestern.edu/tech/dbohn1_table.htm



Single-ended, resistive-loaded Class A stage. 12.5% Maximum efficiency.



Actively-loaded single-ended Class A stage. 25% Maximum efficiency.

<http://diyaudioprojects.com/Solid/IRF610-Class-A-Headphone-Amp/>

Rem. : Haut-parleur : 4 - 8 Ω , >> W



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ELEC1530 – Partie 1 : Bipolaire - 32

$$A_{vo} = \frac{v_o}{v_i} = \frac{R_L}{R_L + r_e} \approx 1$$

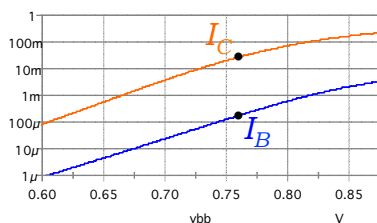
$$R_L = 32 \Omega$$

$$\Rightarrow r_e \approx 1 \Omega$$

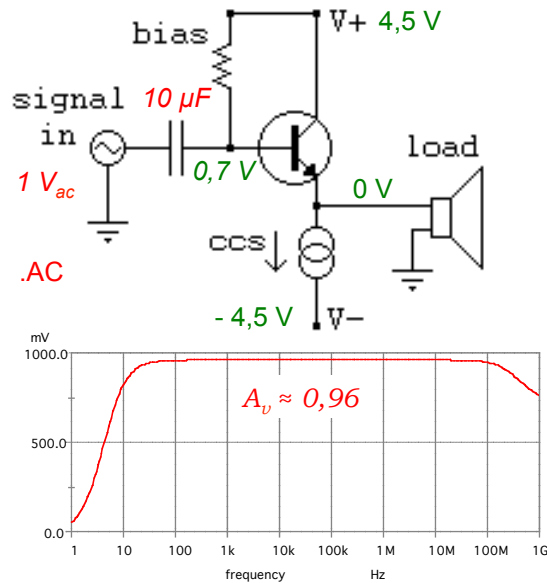
$$r_e = \frac{V_T}{I_E} = \frac{\alpha}{g_m}$$

$$V_T = 26 \text{ mV}$$

$$\Rightarrow I_E \approx I_C \approx 26 \text{ mA}$$

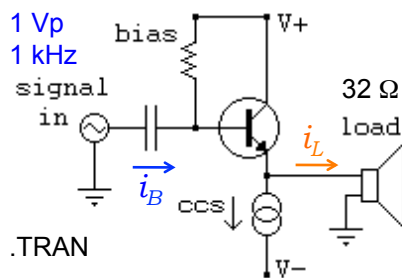


$$I_B \approx 164 \mu\text{A} \Rightarrow R_{bias} = (4,5 - 0,758) / I_B = 23 \text{ k}\Omega \rightarrow .OP$$



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ELEC1530 – Partie 1 : Bipolaire - 33



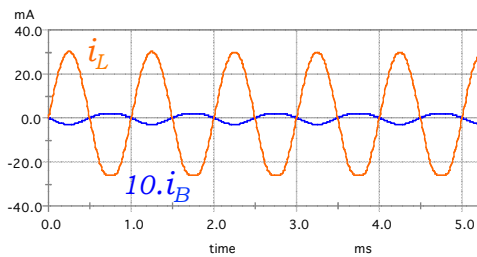
.TRAN

$$P_{eff} = \frac{V_{L,rms}^2}{R_L} = R_L \cdot i_{L,rms}^2 \approx 15 \text{ mW}$$

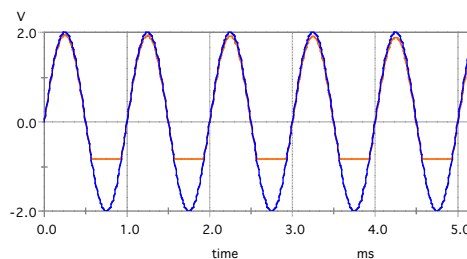
$$P_{dc} = 26 \text{ mA} \cdot 9 \text{ V} \Rightarrow \text{efficacité} = \frac{P_{eff}}{P_{dc}} = 6,4\%$$

in : 2 Vp
out : 'clipping'

$$V_{L,min} = -26 \text{ mA} \cdot 32 \Omega \approx 0,8 \text{ V}$$



$$A_i \approx 110, A_p \approx 105$$



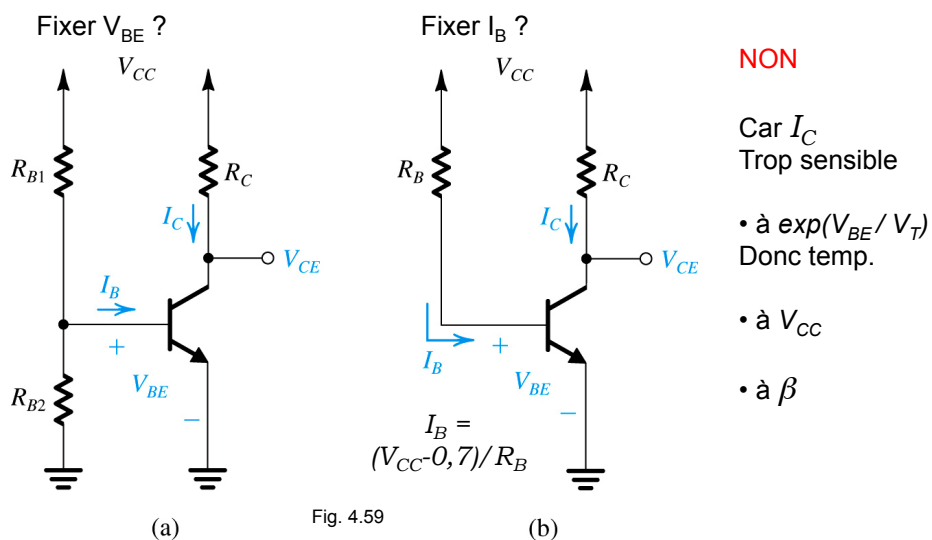
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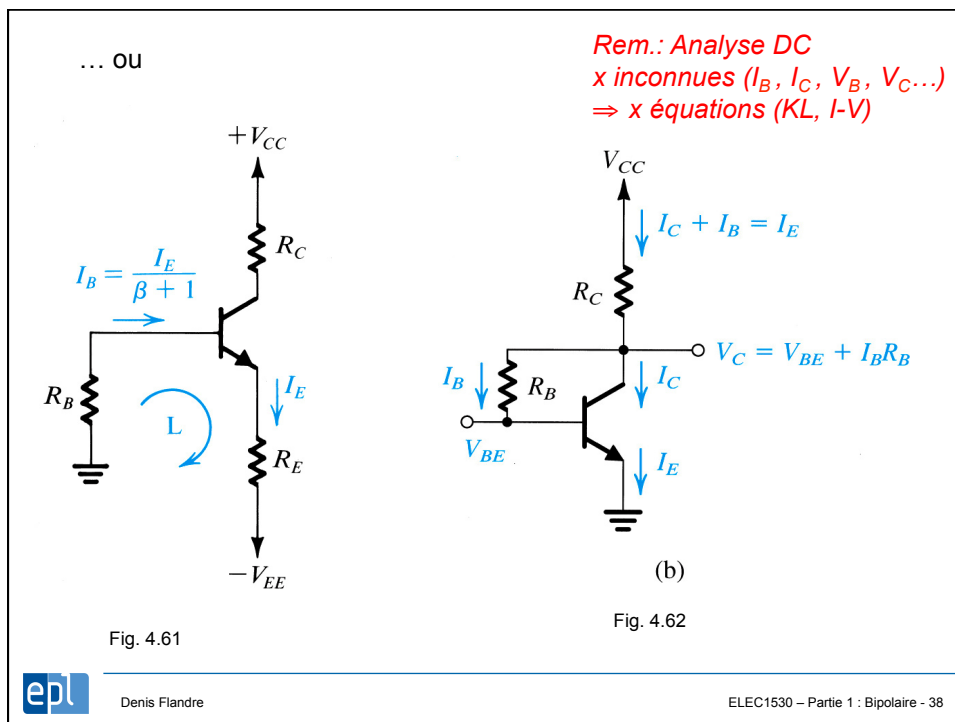
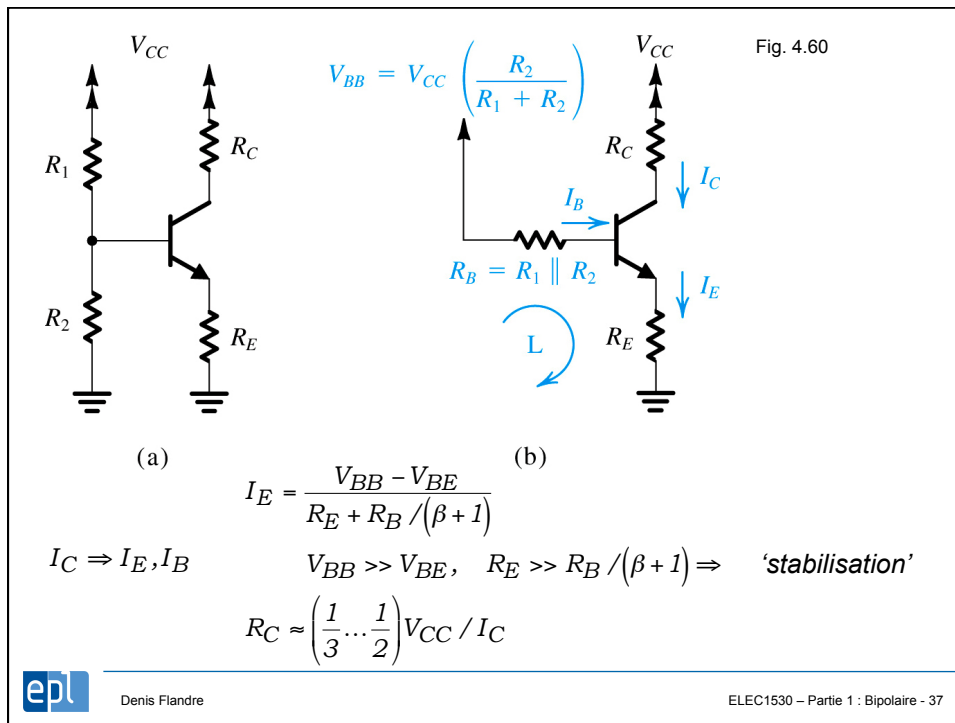
ELEC1530 – Partie 1 : Bipolaire - 34

En résumé

- CE : étage de gain le plus classique (*seul ou en cascade*)
- CE + Re : « *linéarisation* »
- CB : faible R_{in} , mais meilleur en HF que CE
- CC : $A_{vo} \approx 1$
(applications à faible R_L ou suiveur de tension)

4.7 Polarisation des amplificateurs BJT





Polarisation par source de courant

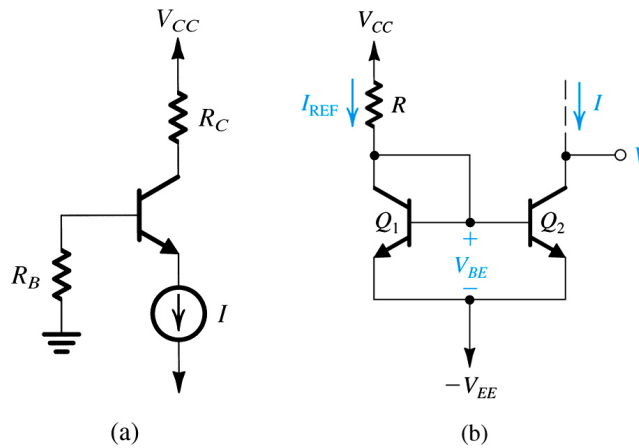


Fig. 4.63

4.8 Amplificateurs BJT discrets

⇒ Configuration (4.6) + Polarisation (4.7) + (Dé)couplage ac (8)

Ex.: émetteur commun

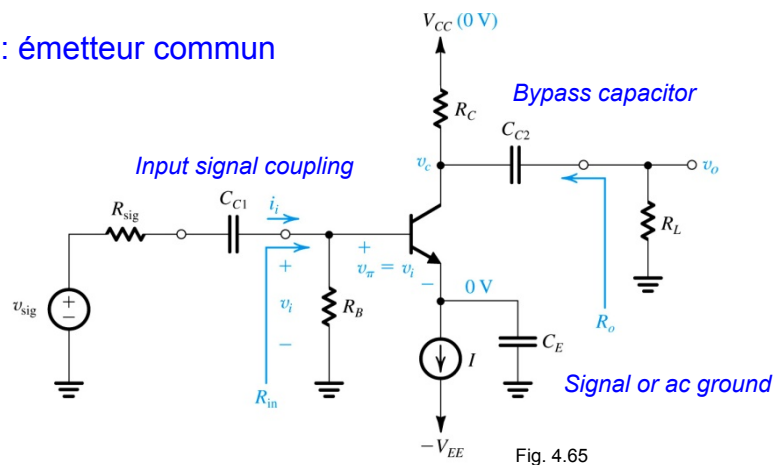


Fig. 4.65

DC ($C = \text{circuits ouverts}$)

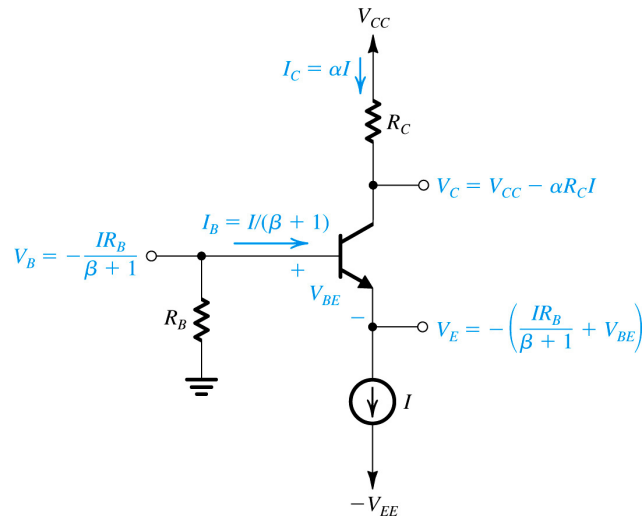


Fig. 4.64

AC (dans la bande passante, $C_{BF} = \text{circuits fermés}$)

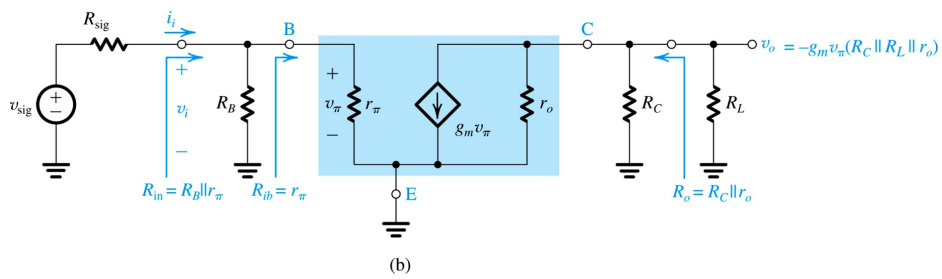


Fig. 4.65

Méthodologie

Dimensionner ce circuit pour une application avec un transistor donné

- Données : MOS (I-V), R_{sig} , R_L , G_v , bande passante
- Inconnues : I_{bias} , R_G , R_D , C

⇒ **Système d'équations interdépendantes** :

- DC : $\forall$ nœuds : KVL, KCL, $V = R \cdot I$ & équations du transistor

$$\bullet \text{ AC : } G_v = - \frac{R_G}{R_G + R_{sig}} g_m \cdot (R_D \parallel R_L \parallel r_o) \cdot \frac{s}{s + \frac{1}{C_{C1} \cdot (R_G + R_{sig})}}$$

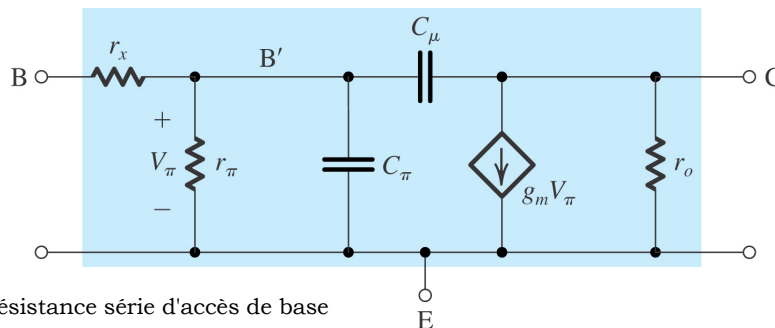
$$\Rightarrow R_G \gg R_{sig}, R_o \ll R_L \text{ \& } (g_m, R_D, I_{bias}), C_{C1} \dots$$

4.9 Effets supplémentaires

- Capacités (8.2.2)

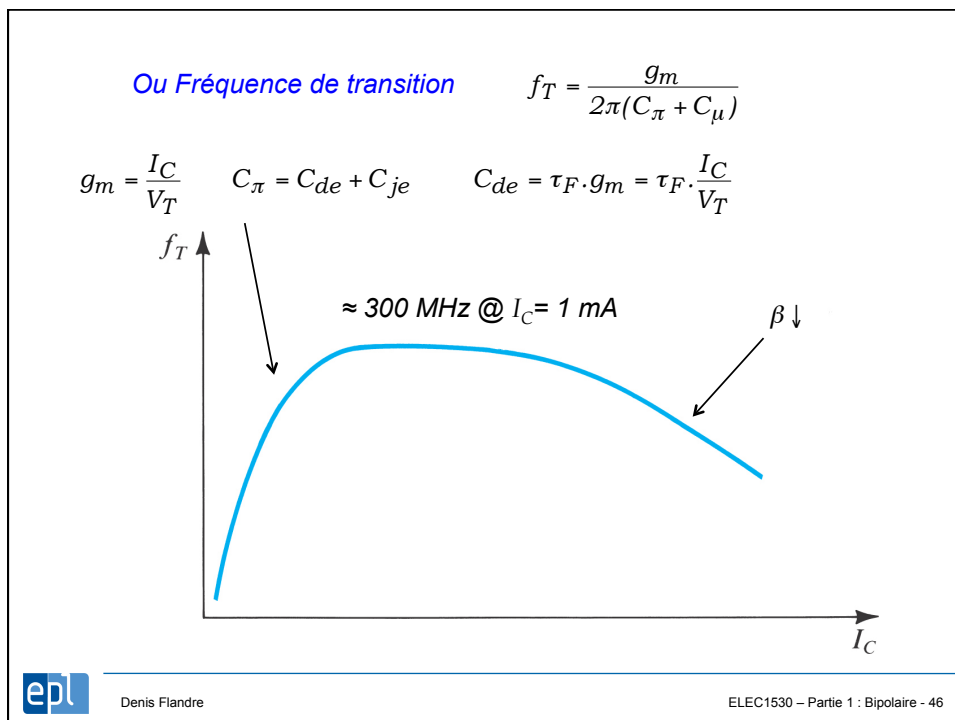
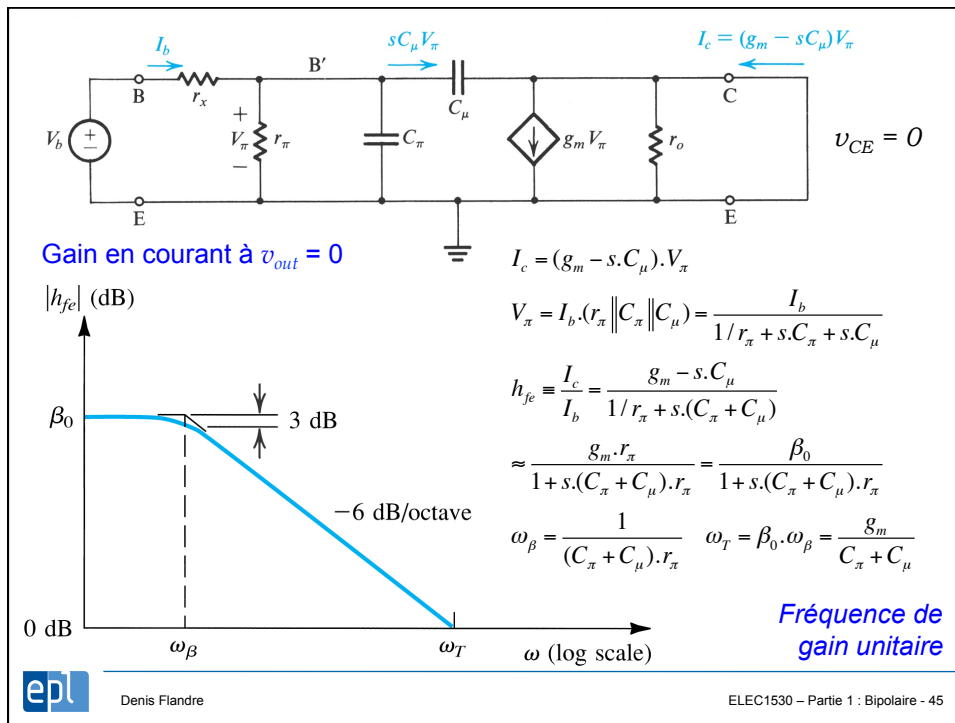
$$C_\pi = C_{de} + C_{je} \quad C_{de} = \tau_F \cdot g_m = \tau_F \cdot \frac{I_C}{V_T} \quad C_{jc} = C_\mu = \frac{C_{\mu 0}}{\left(1 + \frac{V_{CB}}{V_{0c}}\right)^m}$$

C_{je} = capacité de déplétion de la jonction passante BE



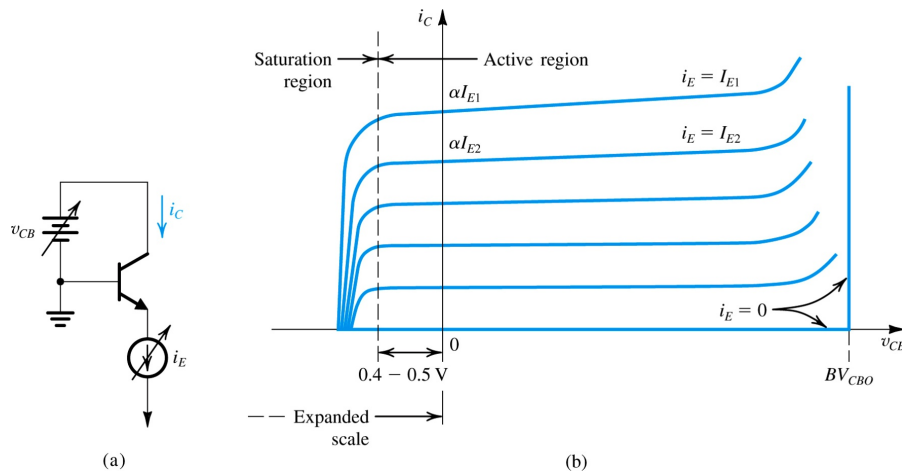
r_x = résistance série d'accès de base
($R_{b(b)}$ dans SPICE)

Modèle π -hybride haute fréquence

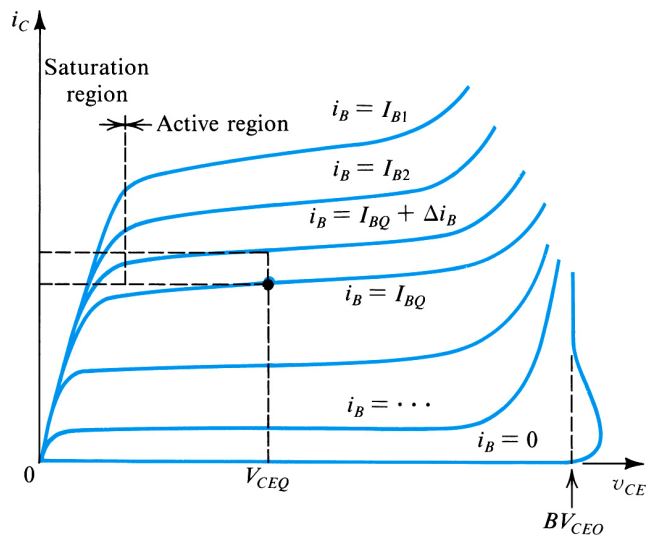


Claquage ou 'Breakdown'

- En base commune



- En émetteur commun



Contenu du chapitre

- 4.1 ~~Structure du transistor BJT~~: *Paramètres du modèle SPICE*
- 4.2 Caractéristiques courant – tension :
 - *Régimes de fonctionnement, Equations I-V*
- 4.3 Circuits BJT DC
- 4.4 Amplificateurs BJT en ‘grand-signal’
 - *Caractéristique de transfert, Point de polarisation, Gain*
- 4.5 Fonctionnement et modèles ‘petit-signal’
 - *Transconductance, Résistance de sortie, Gain, Circuit équivalent*
- 4.6 Configurations d’amplificateurs BJT
 - *CE, CE + Re, CB, CC (suiveur)*
- 4.7 Polarisation : *par V_{BE} , I_B ... ou source de I*
- 4.8 Amplificateurs BJT discrets
- 4.9 Effets supplémentaires : *capacités, réponse en fréquence*, claquage, *température*